

Star-Shaped Nakajima Quiver Varieties, Parabolic Hitchin Moduli Spaces, and their Holomorphic Symplectic Structures

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Star-Shaped Nakajima Quiver Varieties

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Parabolic Higgs Bundles

Let $C = \mathbb{CP}^1$ and $D = \{p_1, \dots, p_n\} \subseteq C$. Let $E \rightarrow C$ be a trivial rank r_* vector bundle. A *parabolic structure* on E is the data of filtrations $F_i^\bullet$ of E_{p_i} along with parabolic weights $\alpha_i^\bullet$:

$$\begin{aligned} E_{p_i} &= F_i^{(\ell_i+1)} \supset \dots \supset F_i^{(1)} \supset 0 \\ \alpha_i^{(\ell_i+1)} &< \dots < \alpha_i^{(1)} \end{aligned}$$

where each $\alpha_i^{(k)} \in (-\frac{1}{2}, \frac{1}{2})$.

These $F_i^\bullet$ may be partial flags, and the number of ‘flag levels’ may depend on $i \in \{1, \dots, n\}$.

Set $r_i^{(k)} = \dim F_i^{(k)}$ and $m_i^{(k)} = r_i^{(k)} - r_i^{(k-1)}$. We assume $\sum_k m_i^{(k)} \alpha_i^{(k)} = 0$.

Parabolic Higgs Bundles

To keep track of the flag types, we fix the *star-shaped quiver* Q and *dimension vector* $\vec{r}$.

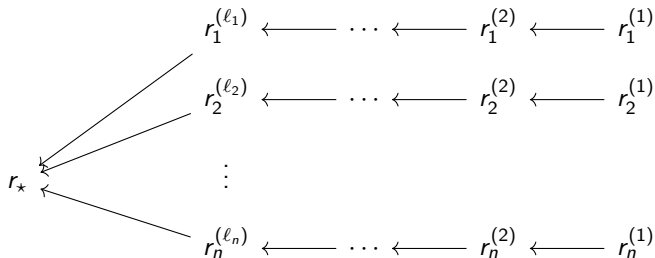


Figure: A star-shaped quiver Q with dimension vector $\vec{r}$

Then the parabolic structure described above is of type $(Q, \vec{r})$.

Parabolic Higgs Bundles

A *parabolic endomorphism* of $\mathcal{E} = (E, \mathcal{F}, \alpha)$ must preserve each $F_i^\bullet$. A *Higgs field* is a trace-free global section φ of $\text{End}_{\text{par}}(\mathcal{E}) \otimes K_C(D)$. We let $M_{(Q, \vec{r}, \mathcal{F})}$ be the space of all pairs $(\bar{\partial}_E, \varphi)$ with $\deg(E, \bar{\partial}_E) = 0$.

Let $\mu_{\mathbb{C}}(\bar{\partial}_E, \varphi) = \bar{\partial}_E \varphi$. Given $(\bar{\partial}_E, \varphi) \in \mu_{\mathbb{C}}^{-1}(0)$, the residue $\varphi_i := \text{res}_{p_i} \varphi$ projects to the Levi (block diagonal) part $\text{Lev}(\varphi_i) \in \mathfrak{l}_i$. This gives a map

$$\mu_{\mathbb{C}}^{-1}(0) \rightarrow \bigoplus_i \mathfrak{l}_i, \quad (\bar{\partial}_E, \varphi) \mapsto \text{Lev}(\bar{\partial}_E, \varphi).$$

If $\text{Lev}(\bar{\partial}_E, \varphi)$ is in the center of $\bigoplus_i \mathfrak{l}_i$, we say $(\bar{\partial}_E, \varphi)$ is a *central-Levi* Higgs bundle.

Fix a $\sigma \in Z(\bigoplus_i \mathfrak{l}_i)$ and define $M_{(Q, \vec{r}, \mathcal{F})}(\sigma) = \text{Lev}^{-1}(\sigma)$.

Parabolic Higgs Bundles

Example (Levi-Central Higgs Fields)

Suppose the i th branch of $(Q, \vec{r})$ is $r_\star = 4 \leftarrow 2 \leftarrow 1$, and $F_i^\bullet$ is $\mathbb{C}^4 \supset \langle e_1, e_2 \rangle \supset \langle e_1 \rangle$. Then at E_{p_i} , any Higgs field φ has the form

$$\varphi_i = \text{res}_{p_i} \varphi = \begin{pmatrix} a_{11} & * & * & * \\ 0 & b_{11} & * & * \\ 0 & 0 & c_{11} & c_{12} \\ 0 & 0 & c_{21} & c_{22} \end{pmatrix} \in \mathfrak{p}_i.$$

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The induced map on the associated graded vector space is given by the triple

$$\text{Lev}(\varphi)_i = \left((a_{11}), (b_{11}), \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \right) \in \mathfrak{l}_i$$

Parabolic Higgs Bundles

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We ask for this to equal some fixed

$$\sigma_i = \left(\sigma_i^{(1)}, \sigma_i^{(2)}, \sigma_i^{(3)} \right) \in Z(\mathfrak{l}_i)$$

We identify $Z(\mathfrak{l}_i) \cong \mathbb{C}^{\ell_i+1}$.

The parabolic weights α determine a notion of parabolic slope stability. Let $\mathcal{G}_{\mathbb{C}} = C^{\infty}(C, \mathrm{SL}(\mathcal{E}))$ be the complex gauge group preserving $\mathcal{F}$ and define

$$\mathcal{M}_{(Q, \vec{r}, \mathcal{F})}(\sigma, \alpha) = \mathrm{Lev}^{-1}(\sigma)^{\alpha-\mathrm{st}} / \mathcal{G}_{\mathbb{C}}.$$

Theorem (Konno '93, Biquard-Boalch '04, Collier-Fredrickson-Wentworth '24)

For generic α , the moduli space $\mathcal{M}_{(Q, \vec{r}, \mathcal{F})}(\sigma, \alpha)$ is a hyperkähler manifold. The holomorphic symplectic form $\Omega_{\mathcal{M}}$ is given by

$$\Omega_{\mathcal{M}}((\dot{A}_1, \dot{\Phi}_1), (\dot{A}_2, \dot{\Phi}_2)) = -\mathbf{i} \int_C \mathrm{tr} \left(\dot{A}_1 \wedge \dot{\Phi}_2 - \dot{A}_2 \wedge \dot{\Phi}_1 \right) \quad (1)$$

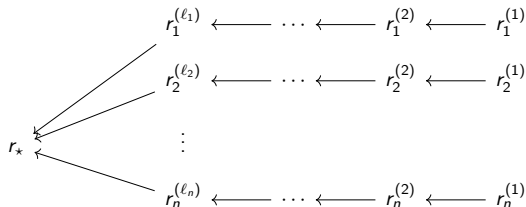
for deformations $(\dot{A}_1, \dot{\Phi}_1), (\dot{A}_2, \dot{\Phi}_2)$ of $(\bar{\partial}_E, \varphi)$.

Given another choice of flags $\mathcal{F}'$, we can conjugate everything by an appropriate $g_{\mathcal{F} \leftarrow \mathcal{F}'} \in \mathfrak{sl}(E)$ to produce an isomorphism $\mathcal{M}_{(Q, \vec{r}, \mathcal{F})}(\sigma, \alpha) \leftrightarrow \mathcal{M}_{(Q, \vec{r}, \mathcal{F}')}(\sigma, \alpha)$. We therefore need not fix $\mathcal{F}$ to define the moduli space $\mathcal{M}_{(Q, \vec{r})}(\sigma, \alpha)$.

However, the formula (1) requires the deformations $(\dot{A}_j, \dot{\Phi}_j)$ to hold the flags fixed. We call these *fixed-flag deformations*.

Star-Shaped Nakajima Quiver Varieties

Recall the data $(Q, \vec{r})$ with $r_i^{(\ell_i)} > \dots > r_i^{(1)}$:



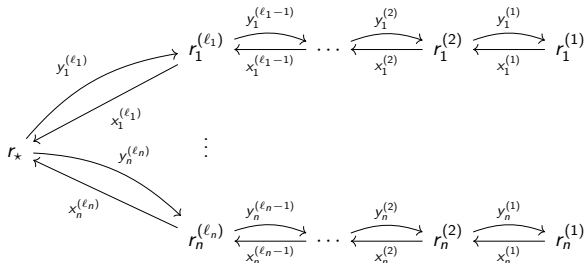
A representation of $(Q, \vec{r})$ is an element

$$\mathbf{x} \in \text{Rep}_{\vec{r}} Q = \bigoplus_{i=1}^n \bigoplus_{k=1}^{\ell_i} \text{Hom} \left(\mathbb{C}^{r_i^{(k)}}, \mathbb{C}^{r_i^{(k+1)}} \right)$$

where $r_i^{(\ell_i+1)} = r_*$.

Star-Shaped Nakajima Quiver Varieties

Define $X = T^* \text{Rep}_{\vec{r}} Q$, which has a natural hyperkähler structure (X, I, ω, Ω) . Then an element of X is a collection of linear maps

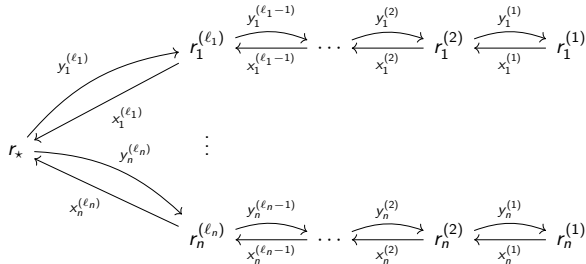


We denote the *doubled quiver* $\tilde{Q}$, so that $X = \text{Rep}_{\vec{r}} \tilde{Q}$. The action of the group

$$G_{\mathbb{C}} = \text{SL}(r_*, \mathbb{C}) \times \bigoplus_{i=1}^n \bigoplus_{k=1}^{\ell_i} \text{GL}(r_i^{(k)}, \mathbb{C})$$

is Hamiltonian with respect to Ω and admits a moment map $\mu_{\mathbb{C}} : X \rightarrow \mathfrak{g}_{\mathbb{C}}^*$.

Star-Shaped Nakajima Quiver Varieties



Since the Lie algebra $\mathfrak{g}_{\mathbb{C}}^*$ is a direct sum, the moment map decomposes

$$\mu_{\mathbb{C}} = \mu_{\mathbb{C},*} + \sum_{i=1}^n \sum_{k=1}^{\ell_i} \mu_{\mathbb{C},i}^{(k)} \text{ where}$$

$$\mu_{\mathbb{C},*}(\mathbf{x}, \mathbf{y}) = - \sum_{i=1}^n \left(x_i^{(\ell_i)} y_i^{(\ell_i)} \right)_0 \in \mathfrak{sl}(r, \mathbb{C})^*$$

$$\mu_{\mathbb{C},i}^{(k)}(\mathbf{x}, \mathbf{y}) = y_i^{(k)} x_i^{(k)} - x_i^{(k-1)} y_i^{(k-1)} \in \mathfrak{gl}(r_i^{(k)}, \mathbb{C})^*$$

The compact real form $G \subset G_{\mathbb{C}}$ is similarly Hamiltonian with respect to ω and admits moment map $\mu_{\mathbb{R}} : X \rightarrow \mathfrak{g}^*$. The moment map decomposes as $\mu_{\mathbb{R}} = \mu_{\mathbb{R},*} + \sum_{i=1}^n \sum_{k=1}^{\ell_i} \mu_{\mathbb{R},i}^{(k)}$ where

$$\mu_{\mathbb{R},*}(\mathbf{x}, \mathbf{y}) = \frac{i}{2} \sum_{i=1}^n \left(-x_i^{(\ell_i)} (x_i^{(\ell_i)})^\dagger + (y_i^{(\ell_i)})^\dagger y_i^{(\ell_i)} \right) \in \mathfrak{su}(r)^*$$

Star-Shaped Nakajima Quiver Varieties

The moment maps $\mu_{\mathbb{C}} : X \rightarrow \mathfrak{g}_{\mathbb{C}}^*$ and $\mu_{\mathbb{R}} : X \rightarrow \mathfrak{g}^*$ are equivariant. Thus, for $\tau \in Z(\mathfrak{g}_{\mathbb{C}}^*) \cong \bigoplus_{i,k} \mathbb{C}$ and $\beta \in Z(\mathfrak{g}^*) \cong \bigoplus_{i,k} \mathbb{R}$, the level sets $\mu_{\mathbb{C}}^{-1}(\tau)$ and $\mu_{\mathbb{R}}^{-1}(\beta)$ are respectively preserved by $G_{\mathbb{C}}$ and G . Restricting to these level sets imposes

$$\mu_{\mathbb{C},\star}(\mathbf{x}, \mathbf{y}) = - \sum_{i=1}^n \left(x_i^{(\ell_i)} y_i^{(\ell_i)} \right)_0 = 0$$

$$\mu_{\mathbb{C},i}^{(k)}(\mathbf{x}, \mathbf{y}) = y_i^{(k)} x_i^{(k)} - x_i^{(k-1)} y_i^{(k-1)} = \tau_i^{(k)}$$

$$\mu_{\mathbb{R},\star}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{i}}{2} \sum_{i=1}^n \left(-x_i^{(\ell_i)} (x_i^{(\ell_i)})^\dagger + (y_i^{(\ell_i)})^\dagger y_i^{(\ell_i)} \right)_0 = 0$$

$$\mu_{\mathbb{R},i}^{(k)}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{i}}{2} \left((x_i^{(k)})^\dagger x_i^{(k)} - y_i^{(k)} (y_i^{(k)})^\dagger - x_i^{(k-1)} (x_i^{(k-1)})^\dagger + (y_i^{(k-1)})^\dagger y_i^{(k-1)} \right) = \beta_i^{(k)}$$

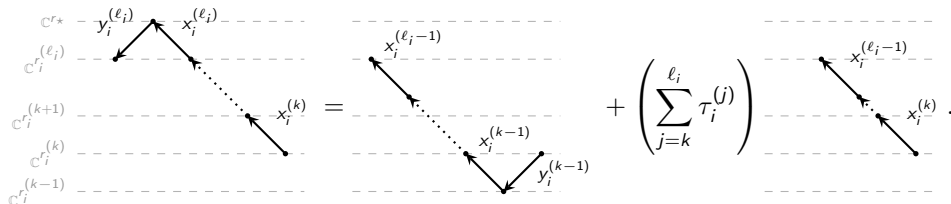
The *Nakajima quiver variety* $\mathcal{X}_{(Q,\vec{r})}(\tau, \beta)$ is $X \mathrel{\mathop{///}\limits_{\tau, \beta}} G = (\mu_{\mathbb{C}}^{-1}(\tau) \cap \mu_{\mathbb{R}}^{-1}(\beta)) / G$.

Star-Shaped Nakajima Quiver Varieties

Lemma

Let $(\mathbf{x}, \mathbf{y}) \in \mu_{\mathbb{C}}^{-1}(\tau)$, and let $X_i^{(k)} = x_i^{(\ell_i)} \cdots x_i^{(k)}$. Then

$$y_i^{(\ell_i)} X_i^{(k)} = X_i^{(k-1)} y_i^{(k-1)} + \left(\sum_{j=k}^{\ell_i} \tau_i^{(j)} \right) x_i^{(\ell_i-1)} \dots x_i^{(k)}.$$



Proof.

We use the relation $\mu_{\mathbb{C},i}^{(k)} = y_i^{(k)} x_i^{(k)} - x_i^{(k-1)} y_i^{(k-1)} = \tau_i^{(k)}$ iteratively to commute the y past the x 's.



Star-Shaped Nakajima Quiver Varieties

By the Kempf-Ness theorem, we have

$$\mathcal{X} //_{\tau, \beta} G \cong \mu_{\mathbb{C}}^{-1}(\tau)^{\beta-\text{st}} / G_{\mathbb{C}}$$

for the following notion of β -stability.

Let $\tilde{\beta} = \left(\beta_{\star}, (\beta_i^{(k)})_{i,k} \right)$ where

$$\beta_{\star} = -\frac{1}{r_{\star}} \sum_{i=1}^n \sum_{k=1}^{\ell_i} r_i^{(k)} \beta_i^{(k)}.$$

For a representation S of $\tilde{Q}$ with dimension vector $\vec{s}$, define

$$\tilde{\beta}(S) = s_{\star} \beta_{\star} + \sum_{i=1}^n \sum_{k=1}^{\ell_i} s_i^{(k)} \beta_i^{(k)}.$$

Theorem (King, '94)

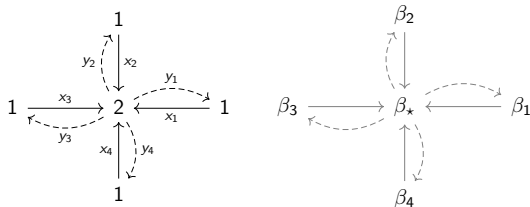
A representation R of Q is β -stable (resp. semistable) if and only if both of the following hold:

- ▶ $\tilde{\beta}(R) = 0$;
- ▶ for all nonzero proper subrepresentations $S \subset R$, we have $\tilde{\beta}(S) < \tilde{\beta}(R)$ (resp. $\tilde{\beta}(S) \leq \tilde{\beta}(R)$).

Star-Shaped Nakajima Quiver Varieties

Example (King Stability)

Consider the data $(\tilde{Q}, \vec{r})$ given by

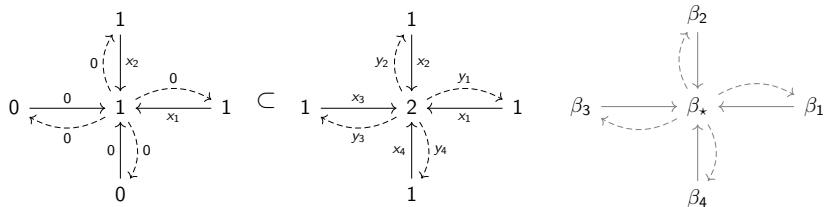


Given weights $\beta_1 > \cdots > \beta_4 > 0$, we extend to $\tilde{\beta}$ by setting $\beta_* = -\frac{1}{2}(\beta_1 + \cdots + \beta_4)$. Suppose $(\mathbf{x}, \mathbf{y}) \in \text{Rep}_{\vec{r}} \tilde{Q}$ has $\mathbf{y} = 0$ and $\text{im } x_1 = \text{im } x_2$.

Star-Shaped Nakajima Quiver Varieties

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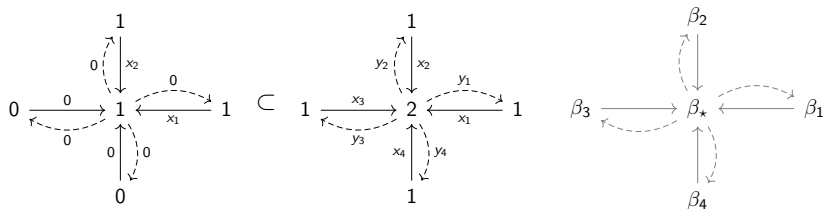


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Star-Shaped Nakajima Quiver Varieties

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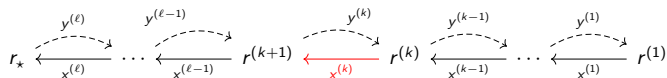
$$\tilde{\beta}(S) = \beta_1 + \beta_2 - \frac{1}{2} \sum_i \beta_i = \frac{1}{2}(\beta_1 + \beta_2 - \beta_3 - \beta_4) > 0.$$

Thus, $(\mathbf{x}, \mathbf{y})$ is not stable.

Star-Shaped Nakajima Quiver Varieties

Example (King Stability)

Consider the data $(\tilde{Q}, \vec{r})$:



Let $\beta^{(1)}, \dots, \beta^{(\ell)} > 0$. Suppose $(\mathbf{x}, \mathbf{y}) \in \text{Rep}_{\vec{r}} \tilde{Q}$ has some $x^{(k)}$ which is not injective.

Star-Shaped Nakajima Quiver Varieties

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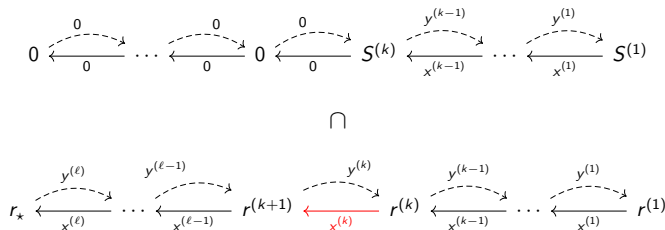
$$\begin{array}{c}
0 \xleftarrow[0]{0} \cdots \xleftarrow[0]{0} 0 \xleftarrow[0]{0} S^{(k)} \xleftarrow[x^{(k-1)}]{y^{(k-1)}} \cdots \xleftarrow[x^{(1)}]{y^{(1)}} S^{(1)} \\
\cap \\
r_{\star} \xleftarrow[x^{(\ell)}]{y^{(\ell)}} \cdots \xleftarrow[x^{(\ell-1)}]{y^{(\ell-1)}} r^{(k+1)} \xleftarrow[x^{(k)}]{y^{(k)}} r^{(k)} \xleftarrow[x^{(k-1)}]{y^{(k-1)}} \cdots \xleftarrow[x^{(1)}]{y^{(1)}} r^{(1)}
\end{array}$$

Let $\beta^{(1)}, \dots, \beta^{(\ell)} > 0$. Suppose $(\mathbf{x}, \mathbf{y}) \in \text{Rep}_{\tilde{r}} \tilde{Q}$ has some $x^{(k)}$ which is not injective. Then $(\mathbf{x}, \mathbf{y})$ admits a subrepresentation S .

Star-Shaped Nakajima Quiver Varieties

Example (King Stability)

Consider the data $(\tilde{Q}, \vec{r})$:



Let $\beta^{(1)}, \dots, \beta^{(\ell)} > 0$. Suppose $(\mathbf{x}, \mathbf{y}) \in \text{Rep}_{\vec{r}} \tilde{Q}$ has some $x^{(k)}$ which is not injective. Then $(\mathbf{x}, \mathbf{y})$ admits a subrepresentation S . Then

$$\tilde{\beta}(S) = \sum_{j=1}^{(k)} s^{(j)} \beta^{(j)} > 0.$$

Thus, $(\mathbf{x}, \mathbf{y})$ is not stable.

The Map from Quiver Representations to Higgs Bundles

This leads to the the construction of a Higgs bundle from a stable $(\mathbf{x}, \mathbf{y}) \in \mu_{\mathbb{C}}^{-1}(\tau)$ introduced by Rayan and Schoposnik. We take the trivial vector bundle $(E, \bar{\partial}) = \mathcal{O}^{\oplus r_\star}$ over $C = \mathbb{CP}^1$ and define the parabolic structures

$$\begin{aligned} E_{p_i} &= F_i^{(\ell_i+1)} \supset \dots \supset F_i^{(1)} \supset 0 \\ \alpha_i^{(\ell_i+1)} &< \dots < \alpha_i^{(1)} \end{aligned}$$

where $F_i^{(k)} = \text{im}(x_i^{(\ell_i)} \dots x_i^{(k)})$. The parabolic weights are uniquely determined by the equations

$$\alpha_i^{(k)} - \alpha_i^{(k+1)} = \beta_i^{(k)}, \quad \sum_{k=1}^{\ell_i} m_i^{(k)} \alpha_i^{(k)} = 0$$

where $m_i^{(k)} = r_i^{(k+1)} - r_i^{(k)}$.

The Map from Quiver Representations to Higgs Bundles

Similarly, determine the complex numbers $\sigma_i^{(k)}$ by the equations

$$\sigma_i^{(k)} - \sigma_i^{(k+1)} = \tau_i^{(k)}, \quad \sum_{k=1}^{\ell_i} m_i^{(k)} \sigma_i^{(k)} = 0.$$

Let $\varphi = \sum \frac{\varphi_i dz}{z - p_i}$ where $\varphi_i = \left(x_i^{(\ell_i)} y_i^{(\ell_i)} \right)_0$.

Lemma

The above construction satisfies the following statements:

1. *The Higgs field φ is holomorphic at infinity.*
2. *The Higgs field is adapted to $(\mathcal{F}, \sigma)$, i.e. $\varphi \in \text{End}_{\text{par}}(\mathcal{E})$ and $\text{Lev}(\bar{\partial}_E, \varphi)_i = \sigma_i$.*

The proof follows from the moment map equations

$$\mu_{\mathbb{C}, \star}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n \left(x_i^{\ell_i} y_i^{(\ell_i)} \right)_0 = 0$$

$$\mu_{\mathbb{C}, i}^{(k)}(\mathbf{x}, \mathbf{y}) = y_i^{(k)} x_i^{(k)} - x_i^{(k-1)} y_i^{(k-1)} = \tau_i^{(k)}$$

Example (Higgs Field Construction)

Suppose the i th branch of $(Q, \vec{r})$ is $r_\star = 4 \leftarrow 2 \leftarrow 1$. Let $(\mathbf{x}, \mathbf{y}) \in \mathcal{X}_{(Q, \vec{r})}(\tau, \beta)$. Say

$$x_i^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_i^{(2)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\mu_{\mathbb{C}, i}^{(k)}(\mathbf{x}, \mathbf{y}) = \tau_i^{(k)}$ for $k = 1, 2$ requires

$$y_i^{(1)} = \begin{pmatrix} \tau_i^{(1)} & a \end{pmatrix}, \quad y_i^{(2)} = \begin{pmatrix} \tau_i^{(1)} + \tau_i^{(2)} & a & * & * \\ 0 & \tau_i^{(2)} & * & * \end{pmatrix}$$

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The parabolic flag is $E_{p_i} \supset \langle e_1, e_2 \rangle \supset \langle e_1 \rangle$, and the Higgs field residue is

$$\varphi_i = \left(x_i^{(2)} y_i^{(2)} \right)_0 = \begin{pmatrix} \tau_i^{(1)} + \tau_i^{(2)} & a & * & * \\ 0 & \tau_i^{(2)} & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} - \frac{1}{4} \left(\tau_i^{(1)} + 2\tau_i^{(2)} \right) \text{Id}.$$

Example (Higgs Field Construction)

$$\begin{aligned}\varphi_i = \left(x_i^{(2)} y_i^{(2)} \right)_0 &= \begin{pmatrix} \tau_i^{(1)} + \tau_i^{(2)} & a & * & * \\ 0 & \tau_i^{(2)} & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} - \frac{1}{4} \left(\tau_i^{(1)} + 2\tau_i^{(2)} \right) \text{Id} \\ &= \frac{1}{4} \begin{pmatrix} \boxed{3\tau_i^{(1)} + 2\tau_i^{(2)}} & * & * & * \\ 0 & \boxed{-\tau_i^{(1)} + 2\tau_i^{(2)}} & * & * \\ 0 & 0 & \boxed{-\tau_i^{(1)} - 2\tau_i^{(2)}} & 0 \\ 0 & 0 & 0 & \boxed{-\tau_i^{(1)} - 2\tau_i^{(2)}} \end{pmatrix}\end{aligned}$$

The Levi part is

$$\text{Lev}(\bar{\partial}_E, \varphi)_i = \frac{1}{4} \left(3\tau_i^{(1)} + 2\tau_i^{(2)}, -\tau_i^{(1)} + 2\tau_i^{(2)}, -\tau_i^{(1)} - 2\tau_i^{(2)} \right) = (\sigma_i^{(1)}, \sigma_i^{(2)}, \sigma_i^{(3)}).$$

Check $\sigma_i^{(k)} - \sigma_i^{(k+1)} = \tau_i^{(k)}$ and $\sum_k m_i^{(k)} \sigma_i^{(k)} = \sigma_i^{(1)} + \sigma_i^{(2)} + 2\sigma_i^{(3)} = 0$.

The Map from Quiver Representations to Higgs Bundles

Theorem (Y '26)

Assume (τ, β) and (σ, α) are related as above, and assume further that $\sum_{i,k} r_i^{(k)} \beta_i^{(k)} < 1$. If $(\mathbf{x}, \mathbf{y}) \in \mu_{\mathbb{C}}^{-1}(\tau)$ is β -stable, then the Higgs bundle $(E, \bar{\partial}_E, \mathcal{F}, \varphi)$ constructed above is α -stable.

We prove this by comparing degree 0 subbundles of $(E, \bar{\partial}_E, \mathcal{F}, \varphi)$ with subrepresentations of $(\mathbf{x}, \mathbf{y})$. The hypothesis $\sum_{i,k} r_i^{(k)} \beta_i^{(k)} < 1$ is necessary to rule out destabilizing subbundles of $(E, \bar{\partial}, \mathcal{F}, \varphi)$ with negative degree.

If $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}', \mathbf{y}')$ are $G_{\mathbb{C}}$ -equivalent, the resulting Higgs bundles are $\mathcal{G}_{\mathbb{C}}$ -equivalent. Thus, we get a map of moduli spaces

$$\mathcal{T} : \mathcal{X}_{(Q, \bar{r})}(\tau, \beta) \rightarrow \mathcal{M}_{(Q, \bar{r})}(\sigma, \alpha).$$

The Map from Quiver Representations to Higgs Bundles

Theorem (Y '26)

The map $\mathcal{T}$ is a homeomorphism onto the locus $\mathcal{M}_0 \subseteq \mathcal{M}_{(Q, \vec{r})}(\sigma, \alpha)$ of Higgs bundles with trivial holomorphic structure.

This $\mathcal{M}_0$ is a Zariski open subset of $\mathcal{M}_{(Q, \vec{r})}(\sigma, \alpha)$.

The dimension of $\mathcal{X}_{(Q, \vec{r})}(\tau, \beta)$ is [Nak94]

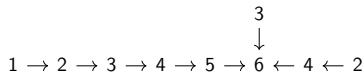
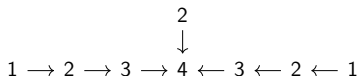
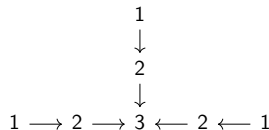
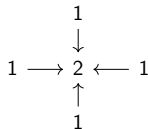
$$\dim_{\mathbb{R}} \mathcal{X}_{(Q, \vec{r})}(\tau, \beta) = 4 \left(1 - r_{\star}^2 + \sum_{i,k} m_i^{(k+1)} r_i^{(k)} \right).$$

The above theorem provides another proof that $\dim_{\mathbb{R}} \mathcal{M}_{(Q, \vec{r})}(\sigma, \alpha)$ is given by the same formula.

$$\dim_{\mathbb{R}} \mathcal{X}_{(Q, \vec{r})}(\tau, \beta) = 4 \left(1 - r_{\star}^2 + \sum_{i,k} m_i^{(k+1)} r_i^{(k)} \right).$$

Example

The following quivers and dimension vectors $(Q, \vec{r})$ produce 4-dimensional Nakajima quiver varieties.



These respectively produce D_4, E_6, E_7, E_8 type ALE gravitational instantons. The corresponding parabolic Higgs bundle moduli spaces are conjecturally ALG instantons.

Holomorphic Symplectic Structures

The holomorphic symplectic forms on $\mathcal{X}_{(Q,\bar{r})}(\tau, \beta)$ and $\mathcal{M}_{(Q,\bar{r})}(\sigma, \alpha)$ are

$$\Omega_{\mathcal{X}}((\dot{\mathbf{x}}_1, \dot{\mathbf{y}}_1), (\dot{\mathbf{x}}_2, \dot{\mathbf{y}}_2)) = \sum_{i=1}^n \sum_{k=1}^{\ell_i} \operatorname{tr} \left(\dot{x}_{1,i}^{(k)} \dot{y}_{2,i}^{(k)} - \dot{x}_{2,i}^{(k)} \dot{y}_{1,i}^{(k)} \right).$$

$$\Omega_{\mathcal{M}}((\dot{A}_1, \dot{\Phi}_1), (\dot{A}_2, \dot{\Phi}_2)) = -\mathbf{i} \int_C \operatorname{tr} \left(\dot{A}_1 \wedge \dot{\Phi}_2 - \dot{A}_2 \wedge \dot{\Phi}_1 \right)$$

Theorem (Y '26)

The map $\mathcal{T} : \mathcal{X}_{(Q,\bar{r})}(\tau, \beta) \rightarrow \mathcal{M}_{(Q,\bar{r})}(\sigma, \alpha)$ satisfies

$$\mathcal{T}^* \Omega_{\mathcal{M}} = 2\pi \Omega_{\mathcal{X}}.$$

Corollary

Pulling back the Hitchin fibration $\mathcal{M} \rightarrow \mathcal{B}$ through $\mathcal{T}$ gives $\mathcal{X}_{(Q,\bar{r})}(\tau, \beta)$ an integrable system structure.

Related Work in the D_4 Case

The strongly parabolic case with the D_4 quiver has been thoroughly studied:

Theorem (Godinho-Mandini '11)

The map $\mathcal{T} : \mathcal{X}(0, \beta) \rightarrow \mathcal{M}(0, \alpha)$ is well-defined.

Theorem (Biswas-Florentino-Godinho-Mandini '15)

The map $\mathcal{T} : \mathcal{X}(0, \beta) \rightarrow \mathcal{M}(0, \alpha)$ preserves the holomorphic symplectic forms.

Theorem (Fredrickson-Y, Heller-Heller-Meneses)

For β in an appropriate chamber, there is a family of hyperkähler metrics on $\mathcal{M}(0, \alpha)$ which converges to the hyperkähler metric on $\mathcal{X}(0, \beta)$.

Theorem (Kronheimer '89)

As (τ, β) varies, the Nakajima quiver varieties $\mathcal{X}(\tau, \beta)$ exhaust all ALE- D_4 instantons.

Theorem (Fredrickson-Mazzeo-Swoboda-Weiss)

Under the assumption that the $\mathcal{M}(\sigma, \alpha)$ really are ALG- D_4 instantons for generic (σ, α) , as (σ, α) varies the spaces $\mathcal{M}(\sigma, \alpha)$ nearly exhausts the entire family of ALG- D_4 instantons.

Proof Strategy

Theorem (Y '26)

The map $\mathcal{T} : \mathcal{X}_{(Q,\bar{r})}(\tau, \beta) \rightarrow \mathcal{M}_{(Q,\bar{r})}(\sigma, \alpha)$ satisfies

$$\mathcal{T}^* \Omega_{\mathcal{M}} = 2\pi \Omega_{\mathcal{X}}.$$

The proof is carried out by pushing forward deformations of $(\mathbf{x}, \mathbf{y})$. Given a family $(\mathbf{x}(t), \mathbf{y}(t)) \in \mu_{\mathbb{C}}^{-1}(\tau) \cap \mu_{\mathbb{R}}^{-1}(\beta)$,

$$(\mathbf{x}(t), \mathbf{y}(t)) \xrightarrow{\mathcal{T}} (\bar{\partial}_E, \mathcal{F}(t), \varphi(t)) \xrightarrow{\text{gauge change}} (\bar{\partial}_{A(t)}, \mathcal{F}(0), \Phi(t))$$

Now the first variation $(\dot{A}, \dot{\Phi})$ is a fixed-flag deformation.

We use this construction to take a pair $(\dot{\mathbf{x}}_1, \dot{\mathbf{y}}_1), (\dot{\mathbf{x}}_2, \dot{\mathbf{y}}_2)$ to a pair $(\dot{A}_1, \dot{\Phi}_1), (\dot{A}_2, \dot{\Phi}_2)$ of fixed-flag deformations, and show that

$$\begin{aligned} \Omega_{\mathcal{M}}((\dot{A}_1, \dot{\Phi}_1), (\dot{A}_2, \dot{\Phi}_2)) &= -\mathbf{i} \int_C \text{tr} \left(\dot{A}_1 \wedge \dot{\Phi}_2 - \dot{A}_2 \wedge \dot{\Phi}_1 \right) \\ &\stackrel{\text{goal}}{=} 2\pi \sum_{i=1}^n \sum_{k=1}^{\ell_i} \text{tr} \left(\dot{x}_{1,i}^{(k)} \dot{y}_{2,i}^{(k)} - \dot{x}_{2,i}^{(k)} \dot{y}_{1,i}^{(k)} \right). \end{aligned}$$

First Variation of Parabolic Flags

We now describe the necessary gauge change to find the fixed-flag deformations $(\dot{A}, \dot{\Phi})$ associated to $(\dot{\mathbf{x}}, \dot{\mathbf{y}})$ —we only need to determine these up to first order in t .

Definition

We say $\dot{\nu} \in C^\infty(C, \text{End}(E))$ induces the first variation of the family $\mathcal{F}(t)$ if for any family of Higgs fields $\varphi(t)$ adapted to $(\mathcal{F}(t), \sigma)$,

$$\left. \frac{d}{dt} \right|_0 e^{-t\dot{\nu}} \varphi(t) e^{t\dot{\nu}} = \dot{\varphi} + [\varphi, \dot{\nu}]$$

is strongly parabolic w.r.t. $\mathcal{F}(0)$.

Then a fixed-flag deformation representing $\mathcal{T}(\mathbf{x}(t), \mathbf{y}(t))$ is

$$(\dot{A}, \dot{\Phi}) = \left. \frac{d}{dt} \right|_0 e^{-t\dot{\nu}} (\bar{\partial}_E, \varphi(t)) e^{t\dot{\nu}} = (-\bar{\partial}_E \dot{\nu}, \dot{\varphi} + [\varphi, \dot{\nu}]) .$$

First Variation of Parabolic Flags

Let $(\mathbf{x}(t), \mathbf{y}(t)) = (\mathbf{x}, \mathbf{y}) + t(\dot{\mathbf{x}}, \dot{\mathbf{y}}) + O(t^2) \in \mu_{\mathbb{C}}^{-1}(\tau) \cap \mu_{\mathbb{R}}^{-1}(\beta)$.

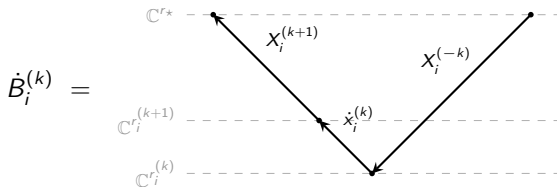
The composition $X_i^{(k)} := x_i^{(\ell_i)} \cdots x_i^{(k)}$ is injective; let $X_i^{(-k)}$ be a left inverse.

Lemma

The variation of the flags is induced by any $\dot{\nu} \in C^\infty(C, \text{End}(E))$ satisfying

$$\dot{\nu}(p_i) = \sum_{k=1}^{\ell_i} \dot{B}_i^{(k)} = \sum_{k=1}^{\ell_i} X_i^{(k+1)} \dot{x}_i^{(k)} X_i^{(-k)}$$

at each puncture $p_i \in D$.

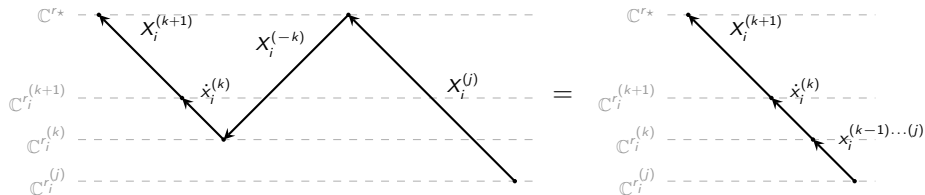


First Variation of Parabolic Flags

This $\dot{B}_i^{(k)}$ satisfies the key property

$$\dot{B}_i^{(k)} X_i^{(j)} = X_i^{(k+1)} \dot{X}_i^{(k)} X_i^{(k-1)} \dots (j)$$

for $j \leq k$.



Proof of Main Theorem (Strongly Parabolic Case)

In the strongly parabolic case where $\sigma, \tau = 0$, the holomorphic symplectic forms $\Omega_{\mathcal{X}}$ and $\Omega_{\mathcal{M}}$ are the exterior derivatives of tautological 1-forms $L_{\mathcal{X}}$ and $L_{\mathcal{M}}$. These are given by

$$L_{\mathcal{X}}|_{(\mathbf{x}, \mathbf{y})}(\dot{\mathbf{x}}, \dot{\mathbf{y}}) = \sum_{i=1}^n \sum_{k=1}^{\ell_i} \operatorname{tr}(\dot{x}_i^{(k)} y_i^{(k)}),$$

$$L_{\mathcal{M}}|_{(\bar{\partial}_E, \varphi)}(\dot{A}, \dot{\Phi}) = -\mathbf{i} \int_C \operatorname{tr}(\dot{A} \wedge \varphi).$$

It suffices to prove $\mathcal{T}^* L_{\mathcal{M}} = 2\pi L_{\mathcal{X}}$. Let $(\dot{A}, \dot{\Phi}) = (-\bar{\partial} \dot{\nu}, \dot{\varphi} + [\varphi, \dot{\nu}])$ be the fixed-flag representative of $d\mathcal{T}(\dot{\mathbf{x}}, \dot{\mathbf{y}})$ from earlier. By Stokes' theorem,

$$L_{\mathcal{M}}|_{(\bar{\partial}_E, \varphi)}(\dot{A}, \dot{\Phi}) = -\mathbf{i} \int_C \operatorname{tr}(\dot{A} \wedge \varphi) = \mathbf{i} \int_C \operatorname{tr}(\bar{\partial} \dot{\nu} \wedge \varphi) = 2\pi \sum_{i=1}^n \operatorname{tr}(\dot{\nu}(p_i) \varphi_i).$$

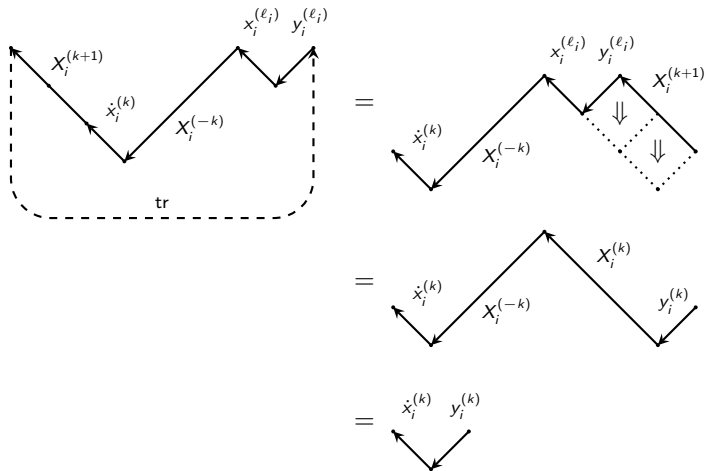
Proof of Main Theorem (Strongly Parabolic Case)

$$L_{\mathcal{M}}|_{(\bar{\partial}_E, \varphi)}(\dot{A}, \dot{\Phi}) = \mathbf{i} \int_C \mathrm{tr}(\bar{\partial} \dot{\nu} \wedge \varphi) = 2\pi \sum_{i=1}^n \mathrm{tr}(\dot{\nu}(p_i) \varphi_i).$$

We have

$$\begin{aligned} \mathrm{tr}(\dot{\nu}(p_i) \varphi_i) &= \sum_{k=1}^{\ell_i} \mathrm{tr}(\dot{B}_i^{(k)} \varphi_i) \\ &= \sum_{k=1}^{\ell_i} \mathrm{tr}(X_i^{(k+1)} \dot{X}_i^{(k)} X_i^{(-k)} x_i^{(\ell_i)} y_i^{(\ell_i)}) \\ &= \sum_{k=1}^{\ell_i} \mathrm{tr}(\dot{X}_i^{(k)} X_i^{(-k)} x_i^{(\ell_i)} y_i^{(\ell_i)} X_i^{(k+1)}) \\ &= \sum_{k=1}^{\ell_i} \mathrm{tr}(\dot{X}_i^{(k)} X_i^{(-k)} X_i^{(k)} y_i^{(k)}) \\ &= \sum_{k=1}^{\ell_i} \mathrm{tr}(\dot{X}_i^{(k)} y_i^{(k)}). \end{aligned}$$

Proof of Main Theorem (Strongly Parabolic Case)



Comments on the Weakly Parabolic Case

We've shown that

$$\mathcal{T}^* L_{\mathcal{M}}(\dot{\mathbf{x}}, \dot{\mathbf{y}}) = L_{\mathcal{M}}(\dot{A}, \dot{\Phi}) = 2\pi \sum_{i=1}^n \sum_{k=1}^{\ell_i} \text{tr}(\dot{x}_i^{(k)} y_i^{(k)}) = 2\pi L_{\mathcal{X}}(\dot{\mathbf{x}}, \dot{\mathbf{y}}),$$

which completes the prove.

The weakly parabolic case is much more complicated, but it boils down to the same idea of using the moment map relations to extract the desired terms from the $\dot{B}_i^{(k)}$.

Thank you!

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